

ECSE 4660/6660: Internetworking of Things

Lecture 1: Course Information & Review of Networking Concepts



Ish Jain
ECSE, RPI

Spring 2026

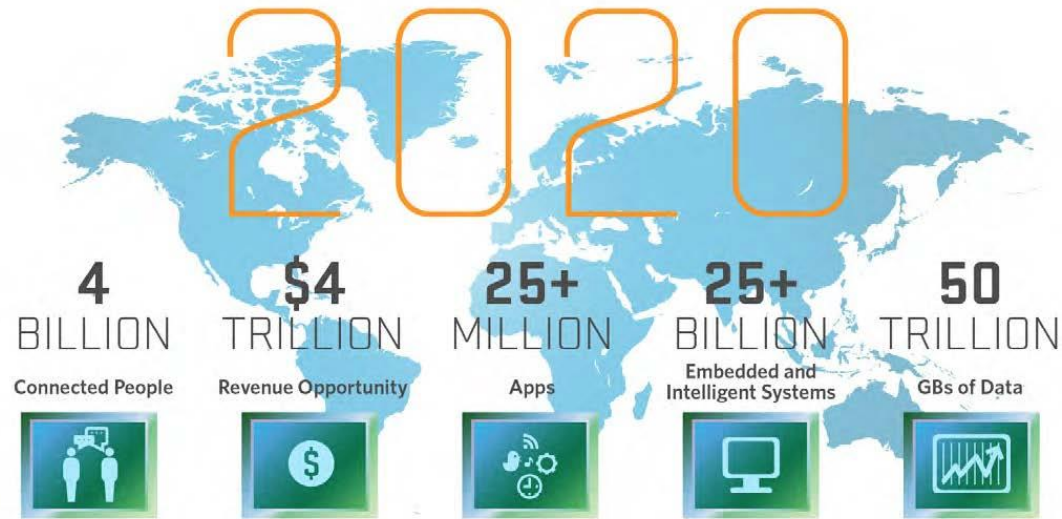
Special Thanks to previous instructors, Koushik Kar and Alhussein Abouzeid

Increasing Demand for Wireless/IoT Connectivity

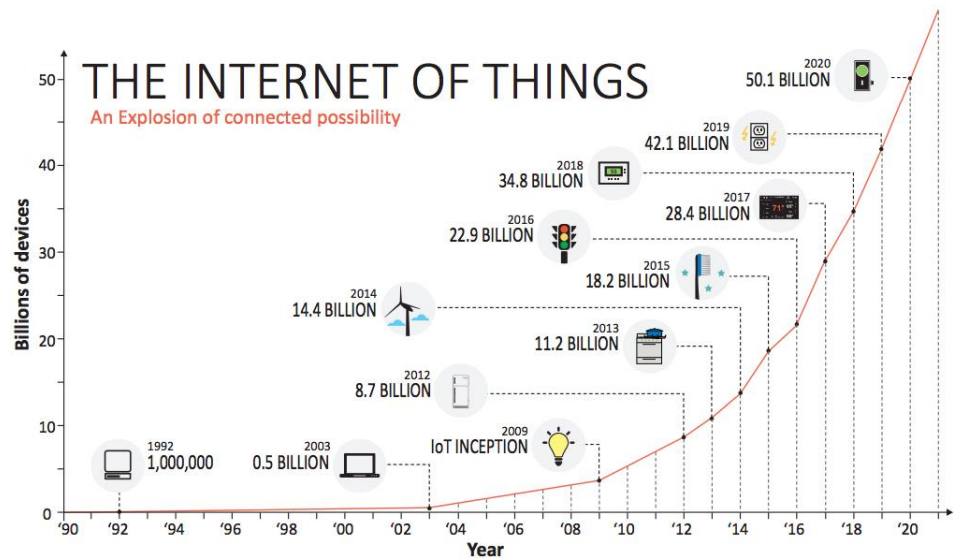
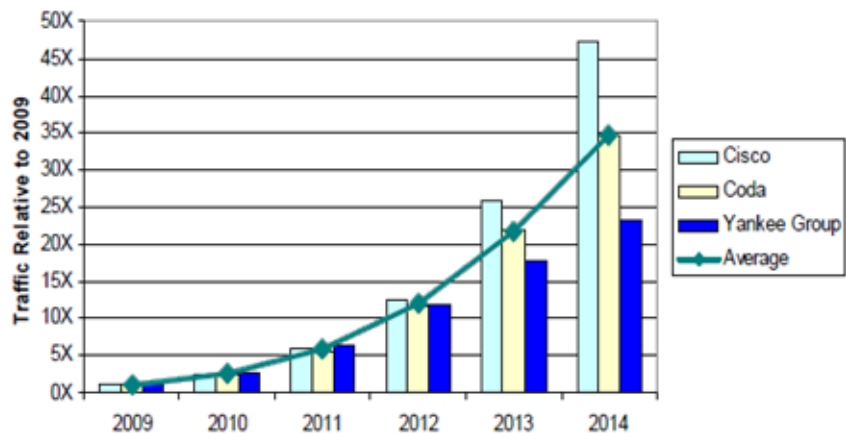


2025?

Increasing Demand for Wireless/IoT Connectivity



Source: Mario Morales, IDC.

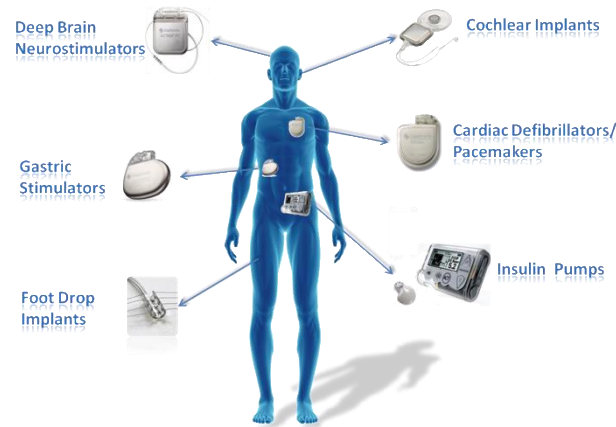


Wireless Sensing and Communication for IoT

Wireless Homes



Wireless Biomedical Implants



Wireless Wearables



Cellular Networks



Wireless Sensors



UAVs



Wireless Data Centers



Wireless VR



Wireless Vehicles



Wireless for Control not just communication

All interaction from objects, machines and humans



Self-driving cars



Robots



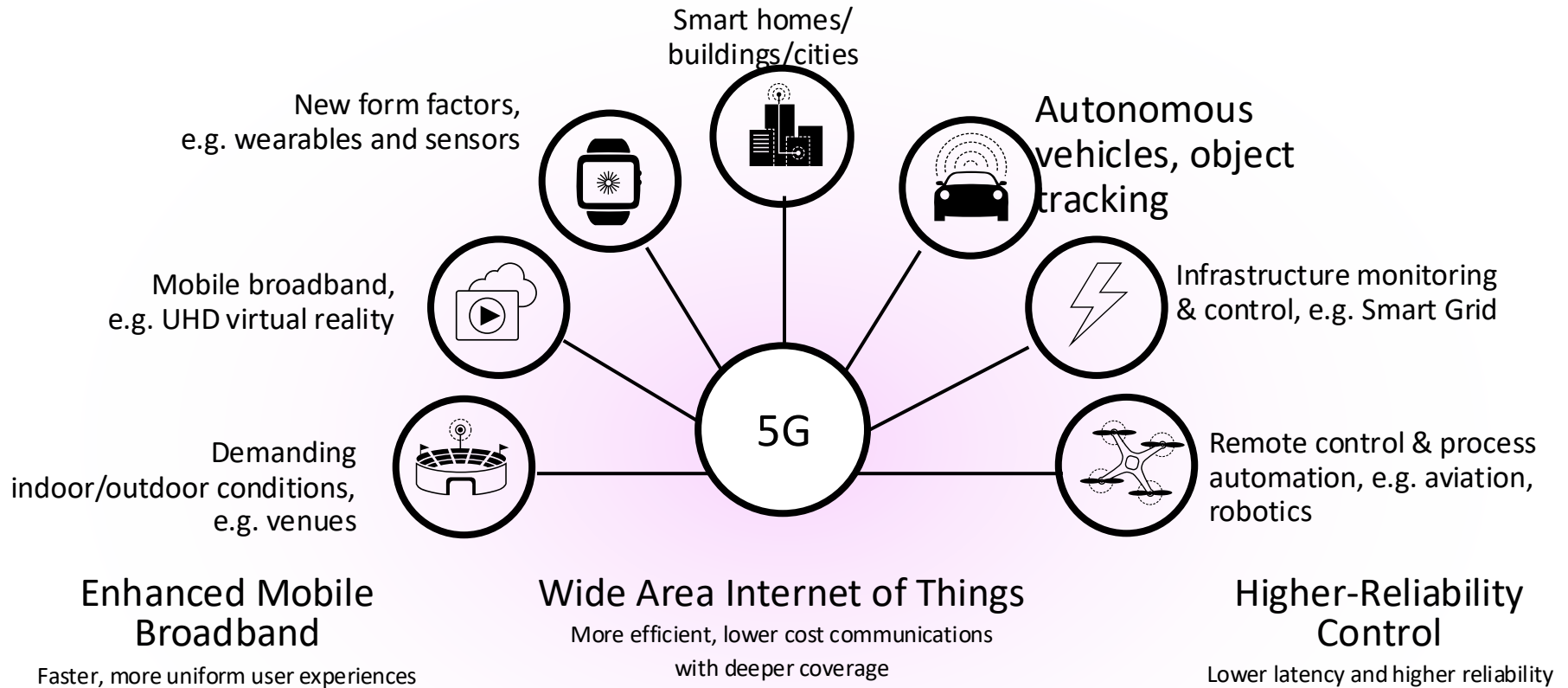
AR, VR



IoT sensors

...a network to accommodate these devices

Pillars of 5G



Source: Qualcomm

ECSE 4660/6660: Internetworking of Things – Spring 2026

ECSE 4560/6560: Modern Communication Systems – Fall 2026

Administrative Information

■ Class meeting

- Lecture Time: M, Th 12:30-1:50 pm
- Location (in-person):
 - JEC 4104
- Lecture on 2/26 will be rescheduled (Hotmobile'26)

■ Instructor

- Professor Ish Jain
 - Office: JEC 6018
 - Email: Jaini@rpi.edu
 - Webex is quicker
- Office Hours:
 - M, Th after class 2-3 PM or by appointment
 - JEC 6018 (office)

■ Lab Times

- In-person:
 - JEC 5312
 - Open Mon-Fri, 7 AM-10 PM
- TA will hold special TA hours in lab
 - TA hours
 - Mon-Fri 4-5 PM
 - JEC 5312

■ TAs

- Chibuikem Ezemaduka
 - ezemac@rpi.edu

Web Resources

- Lab Website: <https://rpi-iot-projects.github.io>
- Piazza “Resources” for Course Material
 - Lecture Slides
 - Reading Lists
 - Lab Assignments
- Piazza “Q&A” for Discussions
 - Annoucements
 - Discussions on Lectures, Labs, HWs, Q&A
- Join Piazza: <https://piazza.com/rpi/spring2026/ecse46606660> (code: internet)
- Webex: DM or project discussion
- GradeScope for assignment submission (you will be added)
- LMS for Quiz and HW submission

Lab and HW Assignments Plan (Tentative)

- There are ~6 labs; and ~6 HWs
- Homeworks will interweave with the Labs
 - Alternate week HWs and Labs
 - Most homeworks will be short answer (T/F, multiple choice, fill in the blank) questions, with no partial credit
- Late Submission Policy (Gradescope)
 - No Late submissions allowed unless approved by the instructor
 - There is a 20% penalty until late submission deadline (typically 2 days later)
 - First two penalties will be waived.
- LMS grading is automatic and doesn't allow late submissions

Lab Plan

■ In-person Labs:

- JEC 5312
- Lab will be open Mon-Fri, 7AM-10PM, accessible with your RPI card
 - Access will be enabled before the first lab assignment is handed out
- Lab help by TAs will be available almost daily
 - Exact schedule is posted on piazza

Lectures and Quizzes

- Attendance
 - While not required, in-person attendance is strongly encouraged. Receive grades for class participation
- Will post a recorded video of each lecture
- A Quiz a day for each lecture
 - Due end-of-day
 - Beyond that, it will be marked late, and will be penalized
 - The quiz will check your knowledge of the material discussed in class
 - Simple questions that can be answered just by attending
 - Questions that require more thinking and studying will be deferred to the homeworks

Course Assessment Measures (tentative)

- Class Participation 5%
- Per-Class Quizzes 5%
- Homework: 20%
- Labwork: 30%
- Final project: 40%

THIS IS A LAB-INTENSIVE COURSE (LABS & PROJECT)

There are no midterm or final exams

Hands-on Final Project

- 2 students per group (need instructor permission if different group size)
- Group formation due in two weeks
 - Proposal due in Week 4, needs to be approved by Instructor
- Prefer students team up with same level (4660 or 6660)
- You may utilize IoT lab resources + you may ask for additional material for project, e.g.,
 - Mobile Robots
 - Drones
 - USRP
 - PlutoSDR
 - Millimeter-wave Radars

Course Material

- IoT devices (available in the lab)
- Optional items (project-dependent)
- No need to buy text-books
 - Selected chapters will be available at the library, linked in each slide set
 - Reading list, which will include
 - Journal and conference articles from IEEE & ACM
 - Supplementary reading list
 - Recommended for interested students
 - Video Recording of each class
 - Lecture Slides
 - Handwritten Notes (if applicable)

Standard Notes

■ Academic Honesty

- Please read about what constitutes academic dishonesty e.g.
 - Failure to cite references
 - Making up results
 - Copying problem solutions/schemes but presenting them differently
 - Presenting work from another course to get credit for multiple courses at once
 - Others
- Penalty for dishonesty
 - Depends on severity and frequency, but could vary including any or all of the following:
 - 0 score on the assignment for 1st instance + reporting to Dean
 - F in the course + reporting to the dean

■ Late Policy

- No late work accepted.
- Special considerations can be given to special situations such as job interview, but must be negotiated *in advance*
- Unexpected events (e.g. sickness or accidents) must be supported by a letter from the dean of students
- With valid excuse, specific assignment weight is removed

Pre-requisites

- Pre-requisites
 - Basic understanding of computer organization, operating systems and networks (CANOS or equivalent)
 - Ability to write computer programs in a high level language (such as C, C++, Python etc.)

Course Goals

- Overall (formal) course goals
 - Technologies and protocols involved in building the Internet-of-Things (IoT)
 - Specific focus on networking at the (wireless) edge of the Internet
- What I really want you to learn
 - IoT is an evolving area, and this course is still undergoing development/changes
 - Basics (lectures) + Application/Reality (lab)
 - Understand how things work, but also why they have been designed that way → design better systems
 - Join the bandwagon of innovation! (through your project)

Course Topics

- Tentative list of core topics to be covered:
 - Review of wireless PHY layer technologies
 - Wireless access technologies for IoT:
 - WiFi, LTE, 5G-NB-IoT,
 - ZigBee, BLE, LoRA, UWB,
 - RFID, Backscatter
 - Routing and network layer protocols for IoT: RPL and 6LoWPAN
 - Low-overhead IoT transport: CoAP and MQTT
 - Network management for IoT: LWM2M and NETCONF
 - IoT mobility models and mobility handling protocols
 - Secure communication of IoT devices over wireless
 - Energy efficient IoT networking
 - Sensor data compression and aggregation

IoT Course Final Projects

- Check out the following for projects from 2023:
<https://www.dropbox.com/scl/fo/4yq8bshbw48yghi6dafk2/h?rlkey=3d32bwf4fygqlyqgs6xtuzrzu&dl=0>
- Projects 2025: <https://rpi-iot-projects.github.io/projects/>
- There should be at least some **communications / networking** element and **hardware** prototype
 - Cannot be entirely sensor data analytics, for example
 - 6660 projects must have a security component
- Most of the projects were suggested by the students themselves
- Projects were done in groups of 2 students
 - Rarely, a project is done by 1 or 3 students, but needs justification/approval

IOT Day 2025

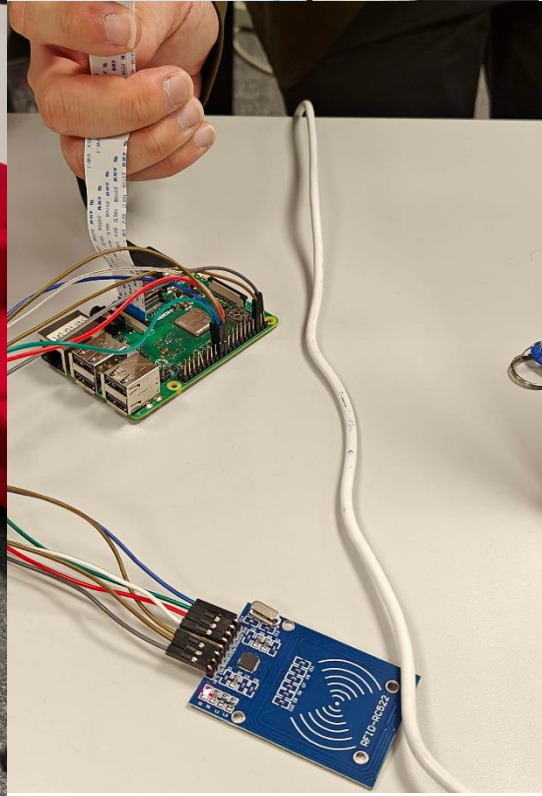
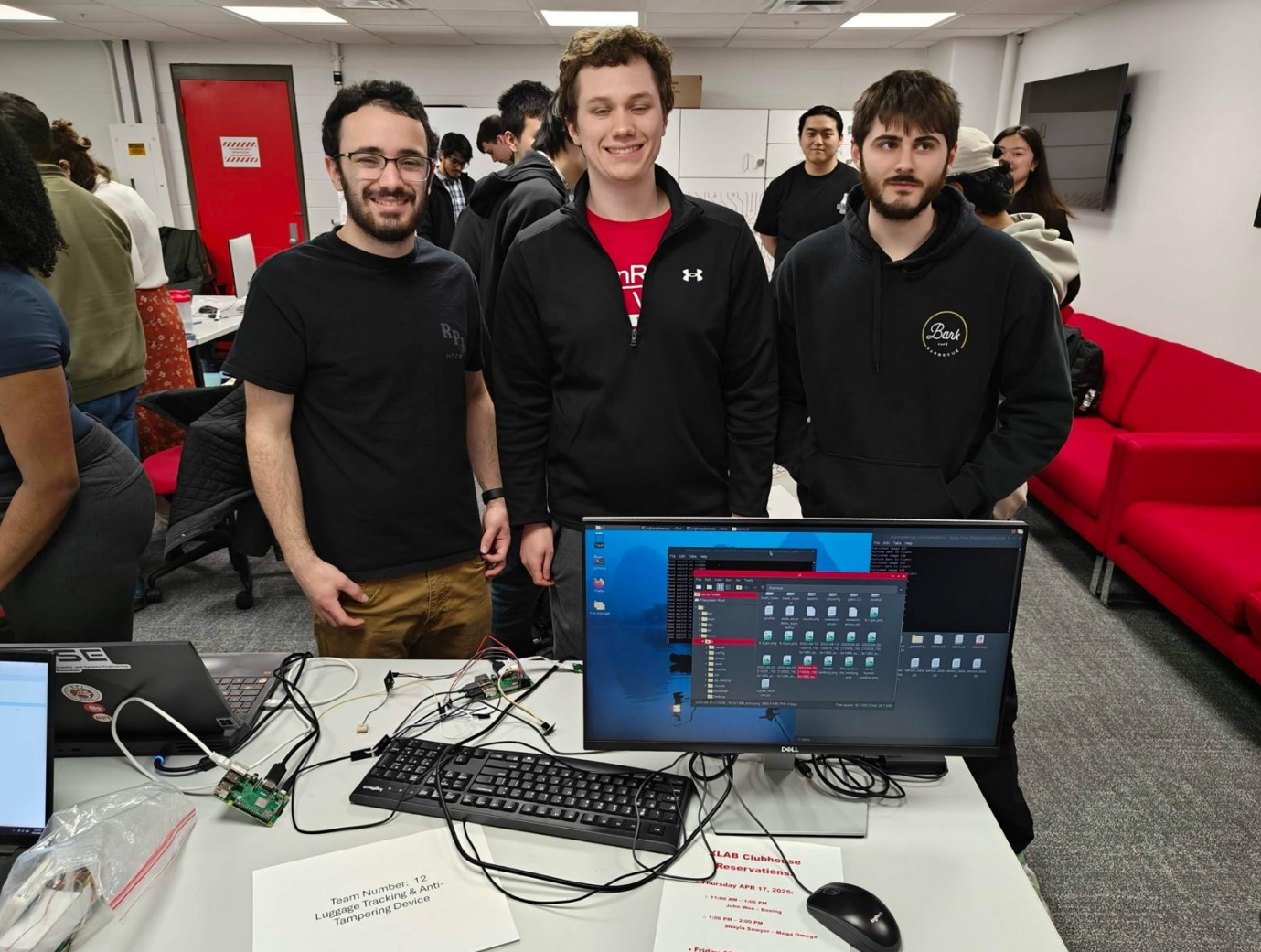


◀ Prev

Pause

Next ▶

IOT Day 2025



Computer Networking: An overview

The Internet: a “nuts and bolts” view



Billions of connected computing *devices*:

- *hosts* = end systems
- running *network apps* at Internet's “edge”



Packet switches: forward packets (chunks of data)

- *routers, switches*

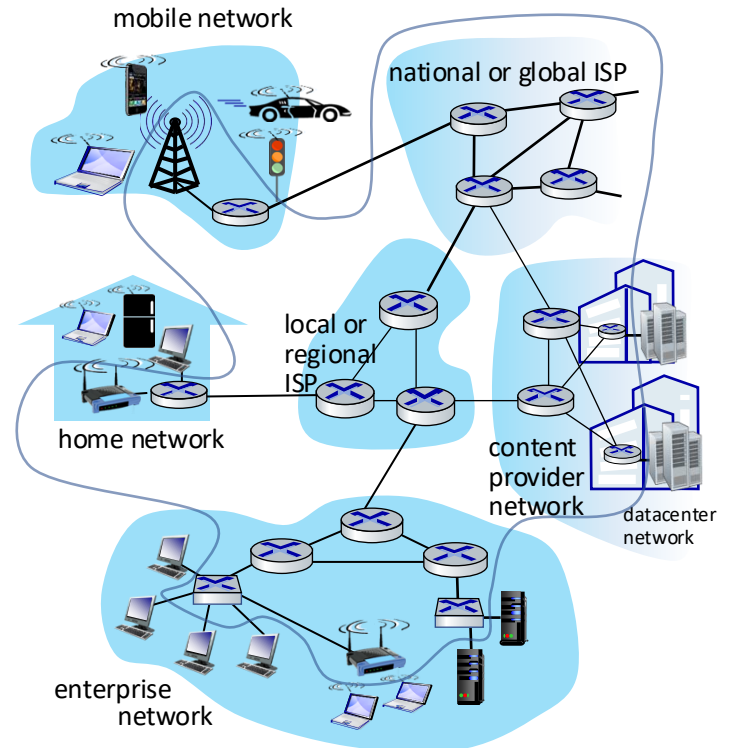


Communication links

- fiber, copper, radio, satellite
- transmission rate: *bandwidth*

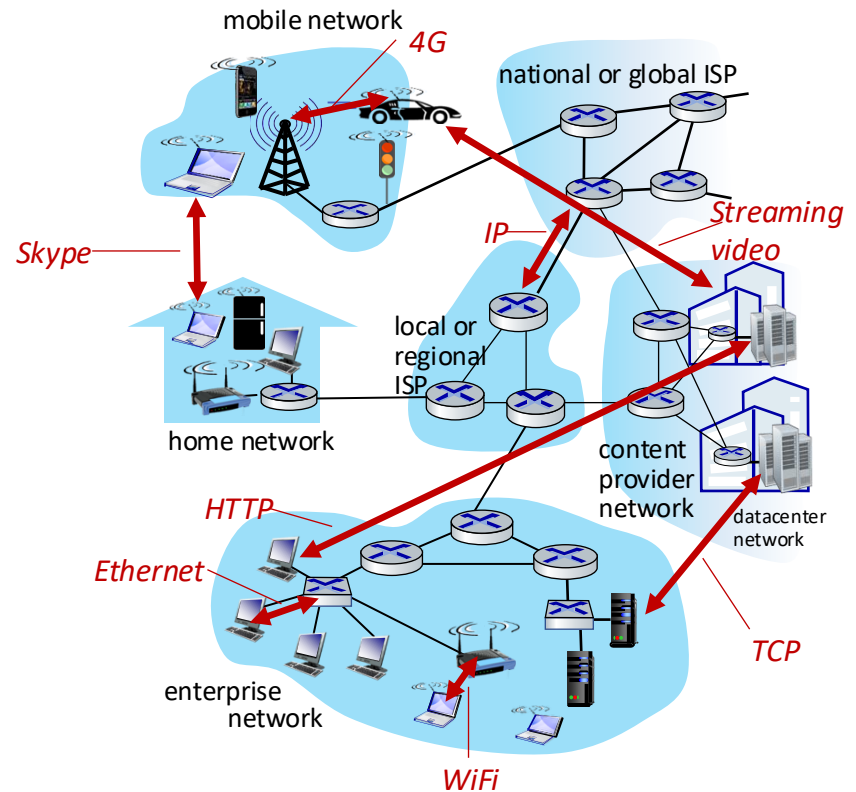
Networks

- collection of devices, routers, links: managed by an organization



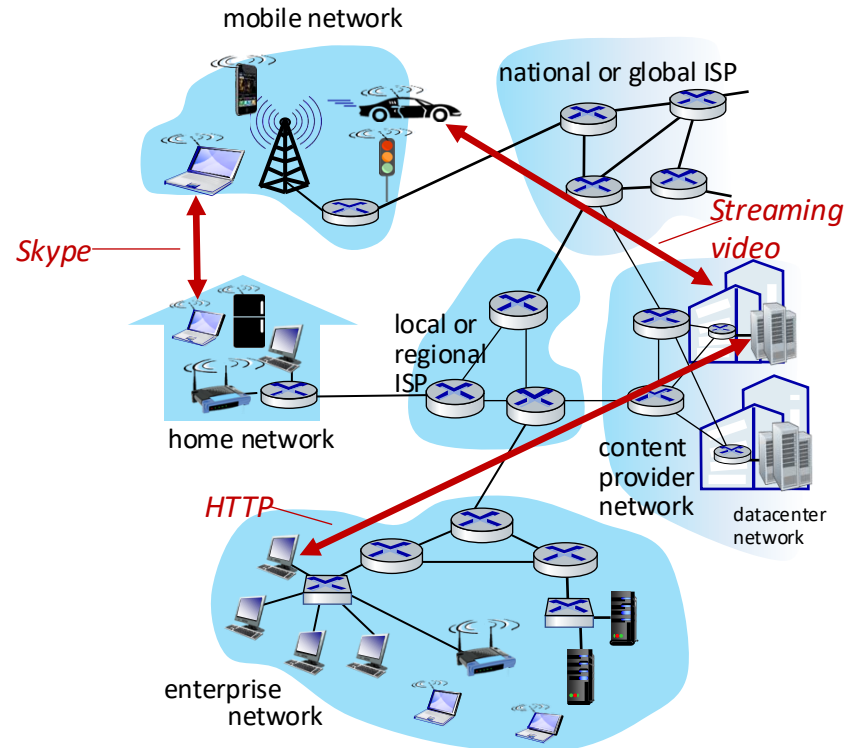
The Internet: a “nuts and bolts” view

- *Internet: “network of networks”*
 - Interconnected ISPs
- *protocols are everywhere*
 - control sending, receiving of messages
 - e.g., HTTP (Web), streaming video, Skype, TCP, IP, WiFi, 4G, Ethernet
- *Internet standards*
 - RFC: Request for Comments
 - IETF: Internet Engineering Task Force



The Internet: a “services” view

- *Infrastructure* that provides services to applications:
 - Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, inter-connected appliances, ...
- provides *programming interface* to distributed applications:
 - “hooks” allowing sending/receiving apps to “connect” to, use Internet transport service
 - provides service options, analogous to postal service



“Fun” Internet-connected devices



Amazon Echo



Internet refrigerator



IP picture frame



Pacemaker & Monitor



Tweet-a-watt:
monitor energy use



Security Camera



Slingbox: remote
control cable TV



Web-enabled toaster +
weather forecaster



AR devices



cars



bikes



scooters



Internet phones



Gaming devices



sensorized,
bed
mattress



Fitbit

Others?

Traditional Approach

Optimize within isolated layers

Network & Apps

Comms. and Coding

HW and Radios

Disruptive gains are unlikely

New Approach

Optimize across the layers

Network and Apps

Comms. and Coding

HW and Radios

Major opportunities!

Cross-Layer Designs

Fundamental Change in Network Architecture

New Services: Wireless Localization

GPS does not work indoor → Use WiFi to localize.



Indoor Navigation



Business Analytics



WiFi Geofencing



Indoor Robotic Navigation

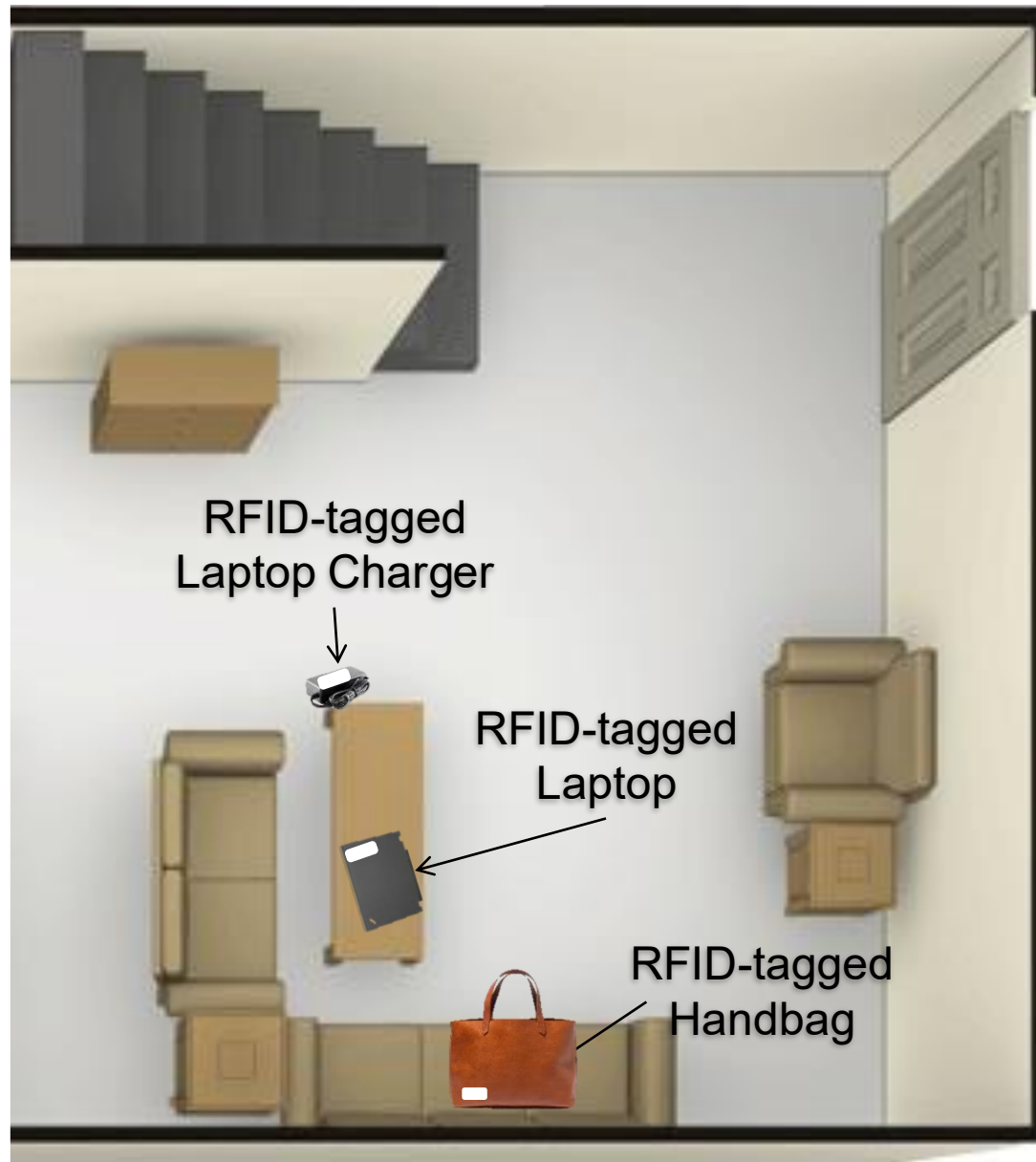
New Services: Wireless IoT Localization

Localize Everything and Anything!



Battery-free stickers to tag any and every object

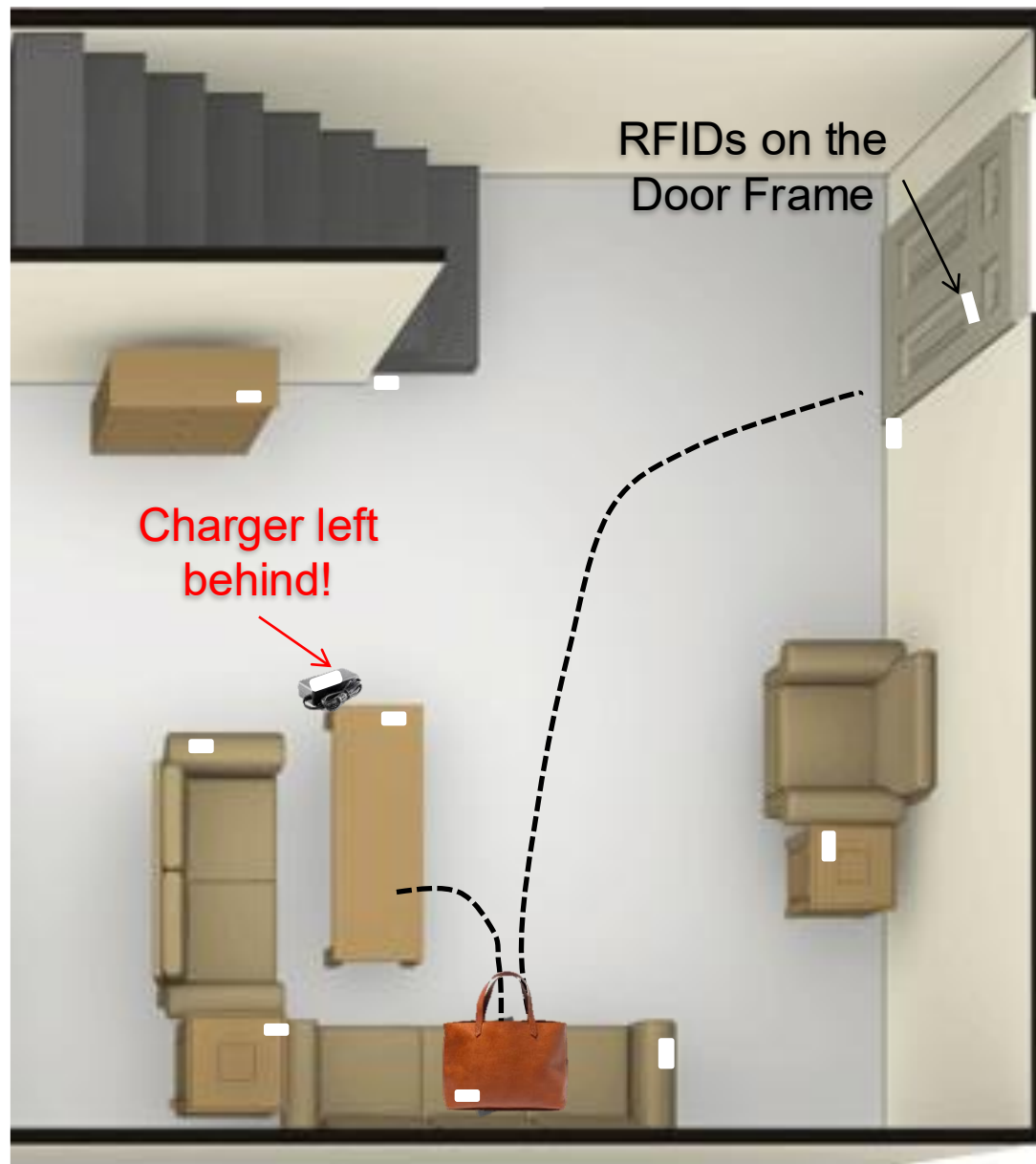
Smart Homes



Smart Homes



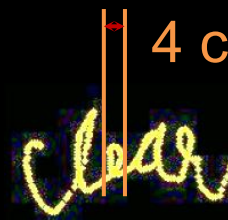
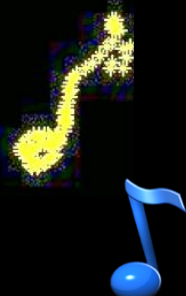
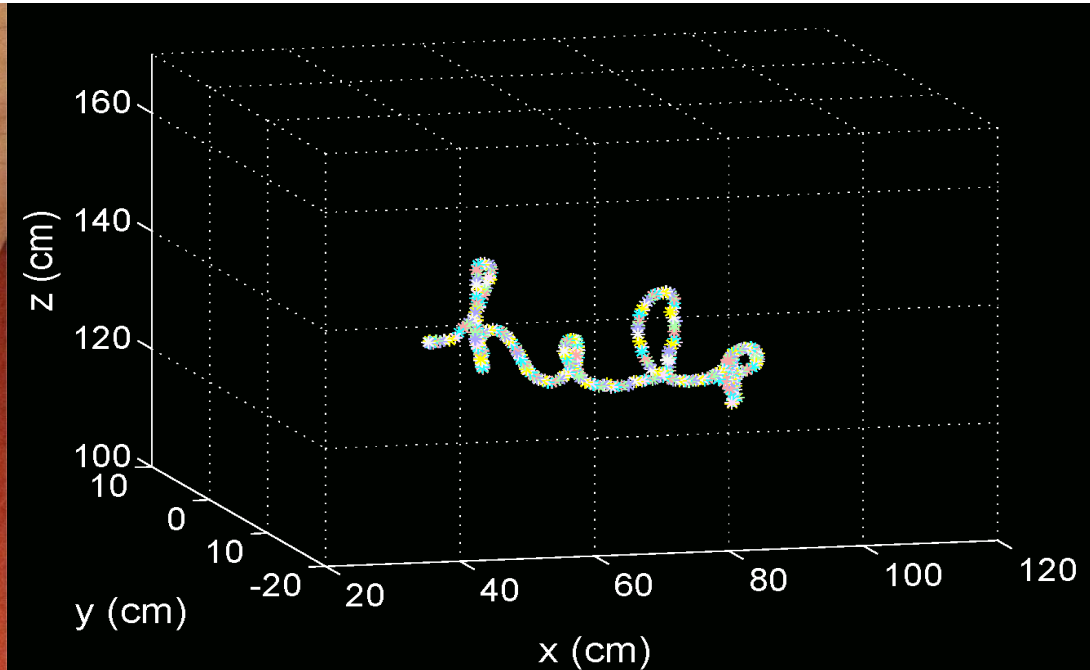
Smart Homes



How Do We Get Virtual Touch Screens?



How Do We Get Virtual Touch Screens?



4 cm wide



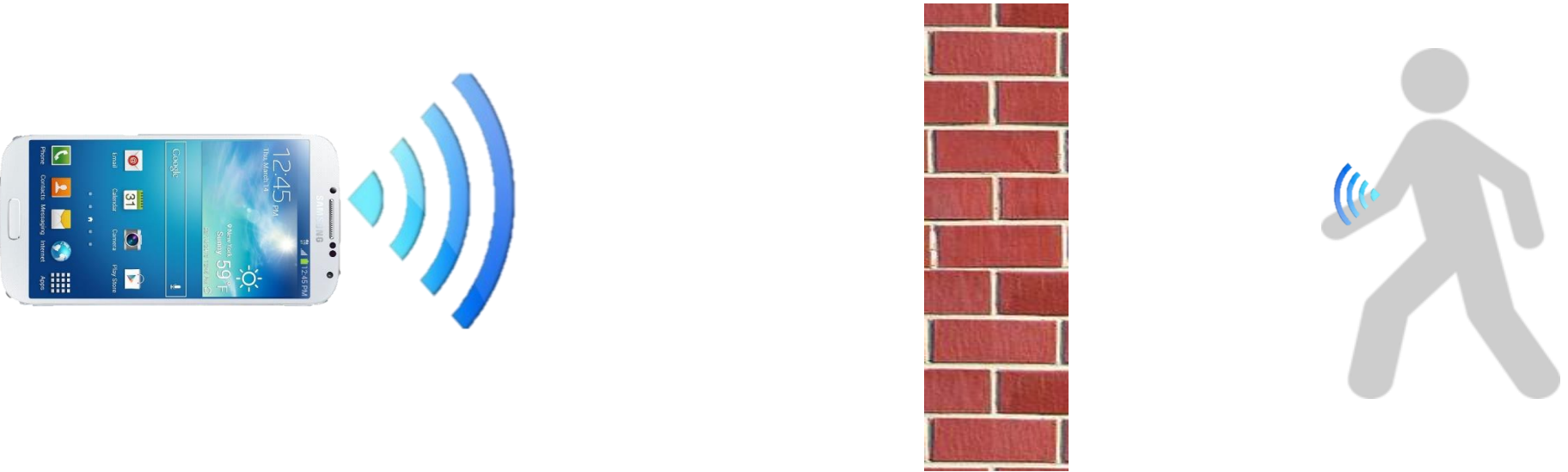
"Clear"

"Jue"

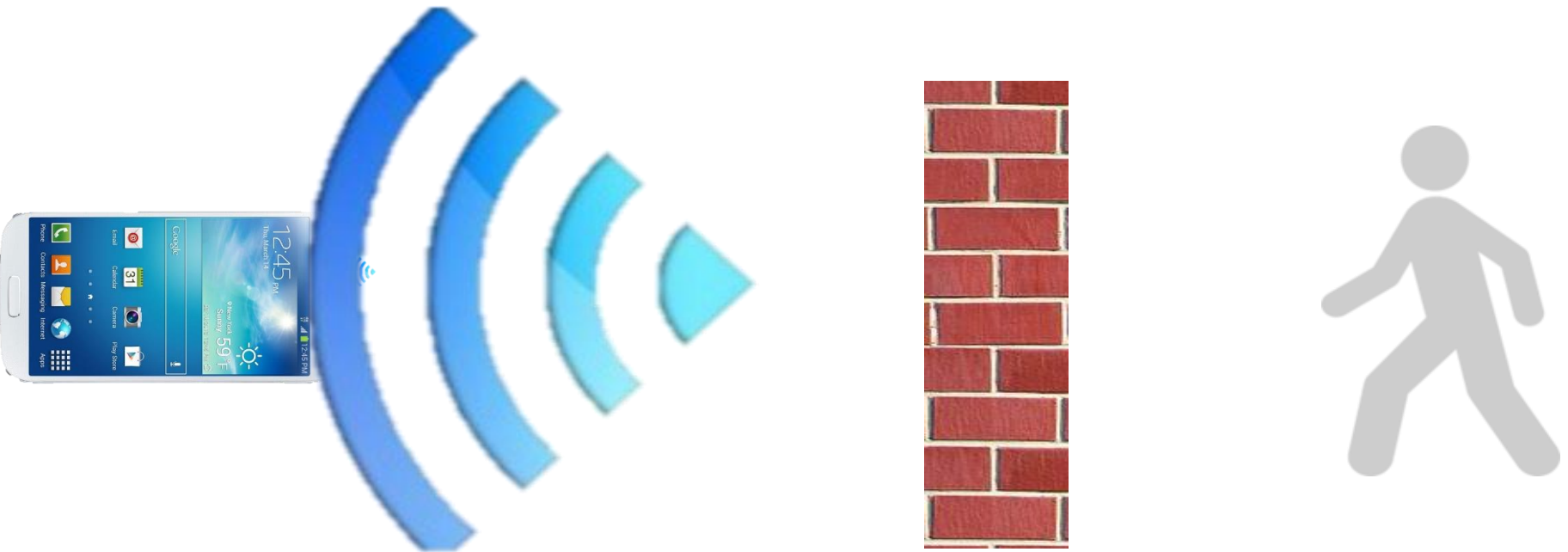
Can your cellphone give you X-ray vision?



See through-walls with WiFi



See through-walls with WiFi

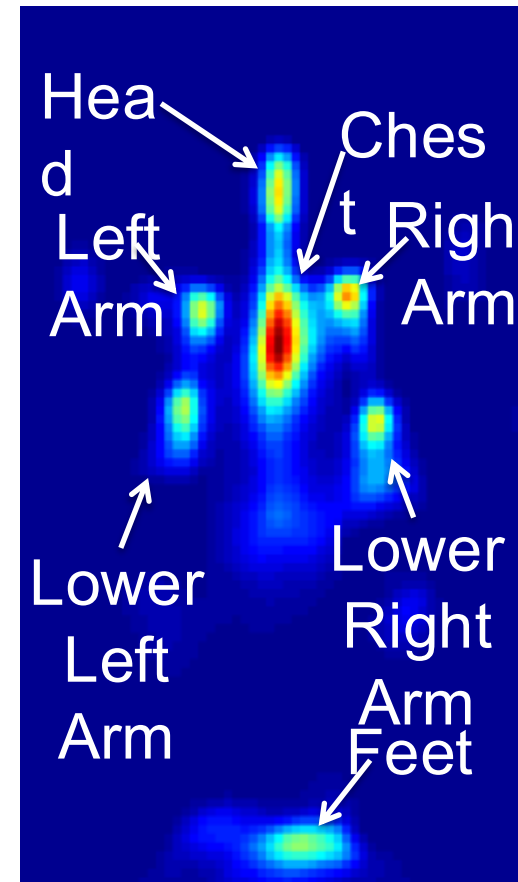
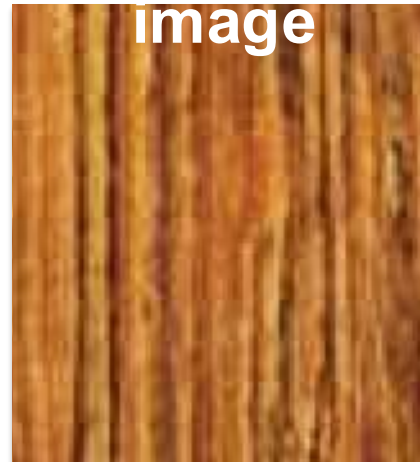


Wall reflection is 10,000x stronger than reflections coming from behind the wall

Human Walks toward Sensor



Imaging through occlusions using radio frequencies



Baby Monitoring

2014-03-14 21:50:30



Baby Monitoring: Breathing and Heart rate



Adib, Fadel, et al. "Smart homes that monitor breathing and heart rate." *Proceedings of the 33rd annual ACM conference on human factors in computing systems*. 2015.

Smart homes that monitor and adapt to our breathing and heart rates?

Personal Health



Baby Sleep



Elderly Health



Adapt Lighting and Music to Mood



Today: technologies for monitoring vital signs are cumbersome

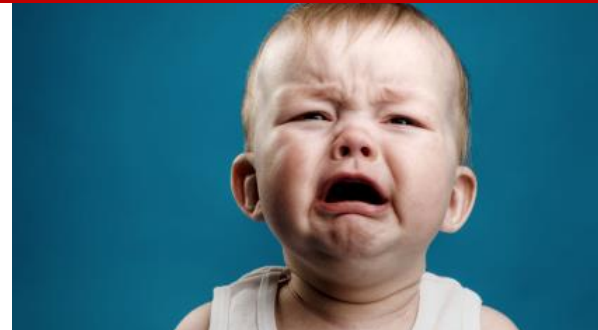
Breath Monitoring



Heart Rate Monitoring



**Wireless enables contactless sensing:
sense humans without any sensors on**



Wireless Security

Medical implants and sensors are power limited

- ➔ Can't have strong cryptography
- ➔ Easy to eavesdrop on the signal, capture confidential data and hack devices.

RFIDs Are Used in Sensitive Applications



Access Control



Credit Cards



Passports



Pharmaceutical Drugs



Anti-Theft Car Immobilizers



Public Transportation

RFIDs Are Used in Sensitive Applications



Access Control

[SECRYPT'09, S&P'09
ESORICS'08,
Usenix'08]



Credit Cards

[DefCon'13,
ShmooCon'12,
DefCon'11 , Usenix'05]



Passports

[DefCon'12, HackaDay'12,
BlackHat'06]



Pharmaceutical Drugs

[CCS'09,
RFID'06]



Anti-Theft Car Immobilizers

[Usenix'12, Usenix'05]



Public Transportation

[Defcon'08, MIT'08,
S&P'09]

Hacking RFIDs for Dummies



www.nicolascourtois.com

www.nicolascourtois.com/MifareClassicHack.pdf

Mifare Classic Tool - MCT

https://play.google.com/store/apps/details?id=de.syss.MifareClassicTool&hl=en

Wirelessly Hack unlock Car

geeknizer.com/wirelessly-hack-unlock-car-without-key-fob/

Live RFID Hacking System

www.openpcd.org/Live_RFID_Hacking_System

Bluetooth Hacking

Open PCD

Navigation Search Toolbox In other languages Views

Page Discussion View source History

Live RFID Hacking System

Bootable RFID Live Hacking System

The bootable Live RFID Hacking System contains a ready-to-use set of hacking tools for breaking and analyzing MIFARE Classic RFID cards and other well known card formats. It is built around PCSC-lite, the CCID free software driver and libirc that gives you access to some of the most common RFID readers. See our tutorial video for a quick introduction on how to break MIFARE Classic RFID card keys using our Live RFID Hacking System.

This RFID Live Hacking System is superseded by our OpenPCD 2 reader with libirc support - you can download the latest ISO image [here](#). This page is only kept for historical reasons.

The MF0C/MF0UK tools of the Live system won't work inside virtualization software like VMware as virtualization seems to break the timing requirements of the MIFARE Classic attack tools - please boot from the CD/DVD instead.

Our RFID hardware projects for RFID Security Analysis

OpenPCD 2 RFID Reader for 13.56MHz
OpenPICC RFID Emulator Project
OpenPICC SnifferOnly 13.56MHz

Suggested RFID Reader for MIFARE Classic key recovery for this live system

Please use the ACR122U102 Tiktag RFID reader for MIFARE key extraction (v1.02) - later versions or compatible models could work, but some later firmware revisions (ACR122U207) seem to be crash while breaking MIFARE Classic with mf0c/mf0c. For normal use and known keys the other compatible readers should be fine though. Please send me a note if you successfully used another reader for key extraction using our Live CD. The Firmware version is shown when using mf0c.

Note for touchatag reader users

If the pcd daemon balls out on a touchatag reader with:

```
00000012 ccid_usb.c:1981:ccid_check_firmware() Firmware (1.00) is bogus! upgrade the reader firmware or get a new reader
00000039 if_handler.c:1181:IFHCreateChannelByname() failed
00000015 readerfactory.c:1998:RFInitialiseReader() Open Port 300000 failed
```

just edit /usr/local/openpcd/lib/pcc/drivers/RF-ccid.bundle/Contents/Info.plist - ifDriverOptions and set key from 0x0000 to 0x0005 to disable version checking.

Checksums

```
Fedora-15-x86_64-Live-RFID-v02.iso
SHA256: 79373eeef8ccbcf2486d4a6356b7f2dd7c86653dbdf2968fce4645db2daa
MD5: c8ef5c1f30b12c30309f9c9e97579de
SHA1: d85409a8959dc8a7668e37618a665897f51ae2
```

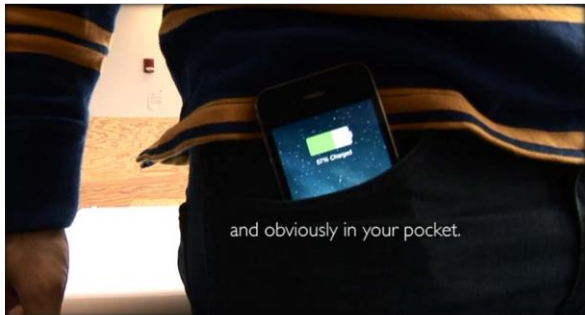
Tools Installed

The most important tools are **highlighted**. The **Fedora 15** based Live Desktop system runs **Gnome 3 Desktop** - just move your mouse cursor in the upper left corner to get a list of installed applications.

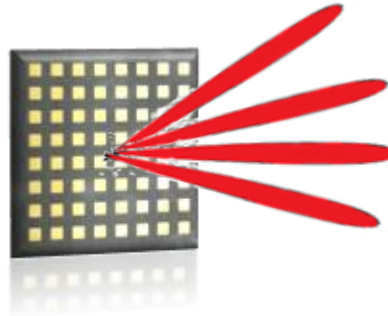
General Purpose Tools

Other Communication Scenarios

Wireless Charging

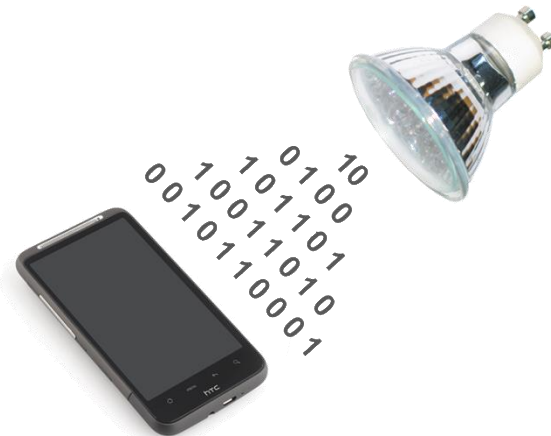


Millimeter Wave Communication



Wireless at Fiber
Optic Speed using
massive antenna
arrays

Visible Light Communication



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